



Optimizing Profitability in Olive Oil Distribution: A Case Study on Viola's Supply Chain Strategy

DECISION THEORY & MULTI-CRITERIA ANALYSIS

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VIOLA OIL CASE STUDY

The Italian oil mill company Viola needs to organize operations beginning from its production facility in Venosa.

- **DISTRIBUTION DETAILS:**

- **Customers:** 10 cities in Southern Italy
- **Fleet:** 3 vehicles (3,500 liters/vehicle)
- **Production capacity:** 8,000 liters/month
- **Frequency:** Monthly deliveries

VIOLA OIL CASE STUDY

- **PRICING SCENARIOS:**

- **Initial price:** €45.2/liter
- **Scenario 1:** €43.5/liter
- **Scenario 2:** €46.4/liter
- **Scenario 3:** €48.2/liter

- **OBJECTIVE FUNCTION**, maximize the profit by optimizing:

- Price uncertainty
- Production and transportation costs
- Delivery routes (distance-based costs)

CVRP

Capacitated Vehicle Routing Problem

CVRP

MATHEMATICAL FORMULATION

The CVRP involves determining optimal routes for a fleet of vehicles to serve a set of customers, such that:

1. The total travel distance is minimized.
2. Each customer's demand is met.
3. The vehicle capacity constraints are not violated.

INDEX:

- I : set of all nodes (depot $i=1$ and customers $i \in \{2, \dots, |I|\}$).
- k : set of vehicles.
- $i, j \in I$: nodes in the network.

CVRP

PARAMETERS:

- d_{ij} : distance between nodes i and j .
- q_i : demand at customer i (depot has no demand).
- B : vehicle capacity.

DECISION VARIABLES:

- $x_{ijk} \in \{0, 1\}$: binary variable indicating whether vehicle k travels from node i to node j .
- $P_{ik} \geq 0$: auxiliary variables for subtour elimination, Miller-Tucker-Zemlin (MTZ) method.

CVRP

OBJECTIVE FUNCTION:

$$\text{Minimize } \sum_{k \in K} \sum_{i \in I} \sum_{j \in I, i \neq j} d_{ij} x_{ijk}$$

SUBJECT TO:

- Flow Conservation:

$$\sum_{i \in I, i \neq j} x_{ijk} = \sum_{i \in I, i \neq j} x_{jik}, \quad \forall j \in I \setminus \{1\}, k \in K$$

- Visit Each Customer Exactly Once:

$$\sum_{k \in K} \sum_{i \in I, i \neq j} x_{ijk} = 1, \quad \forall j \in I \setminus \{1\}$$

- Depot Departure:

$$\sum_{j \in I \setminus \{1\}} x_{1jk} = 1, \quad \forall k \in K$$

- Vehicle Capacity:

$$\sum_{i \in I \setminus \{1\}} \sum_{j \in I, j \neq i} q_i x_{ijk} \leq B, \quad \forall k \in K$$

- Subtour Elimination (MTZ):

$$P_{ik} - P_{jk} + |I| x_{ijk} \leq |I| - 1, \quad \forall i, j \in I \setminus \{1\}, i \neq j, k \in K$$

$$P_{ik} \geq 0, \quad P_{1k} = 0, \quad P_{ik} \leq |I| - 1, \quad \forall i \in I, k \in K$$

ROUTES

ROUTE 1:

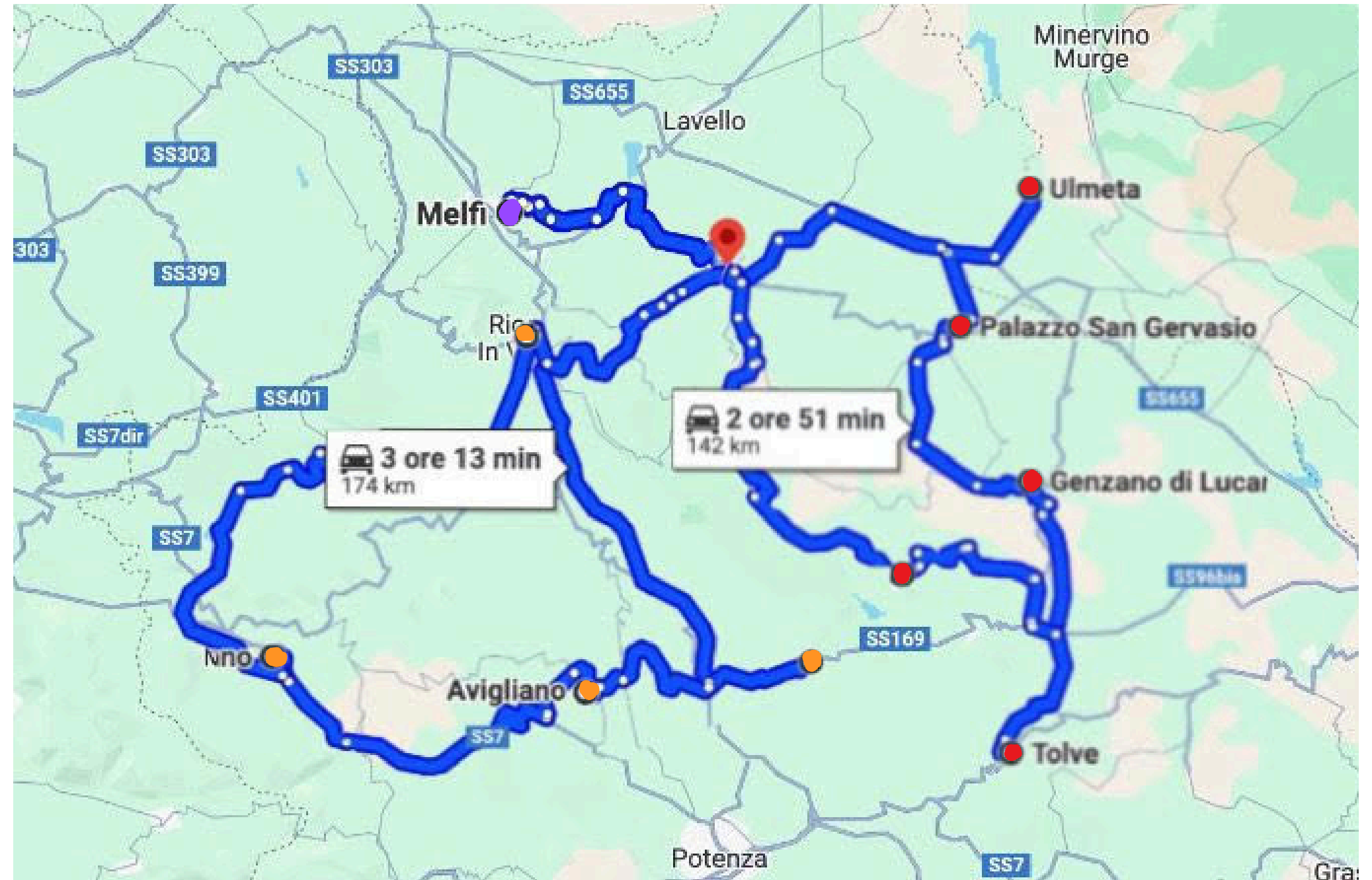
Venosa → Ulmeta → Palazzo San Gervasio → Genzano di Lucania → Tolve → Acerenza → Venosa

ROUTE 2:

Venosa → Pietragalla → Avigliano → Muro Lucano → Rionero In Vulture → Venosa

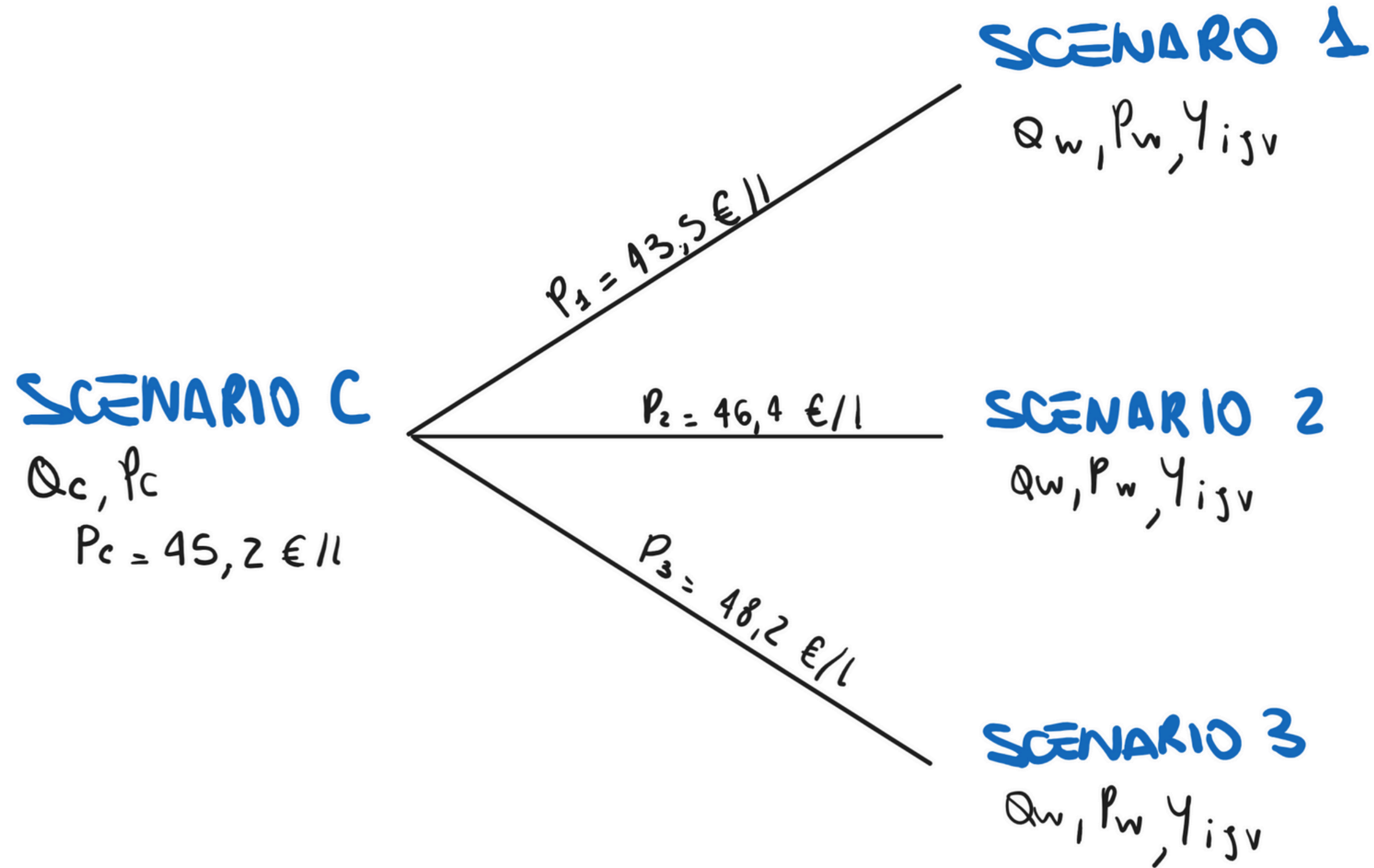
ROUTE 3:

Venosa → Melfi → Venosa



TWO STAGE STOCHASTIC PROBLEM

SCENARIO TREE



MATHEMATICAL FORMULATION

INDEX:

- **i**: cities, starting point $i = 1, \dots, 11$
- **j**: cities, delivering point $j = 1, \dots, 11$
- **v**: vehicles $v = 1, \dots, 3$
- **w**: scenarios $w = 1, \dots, 3$

MATHEMATICAL FORMULATION

PARAMETERS:

- **C**: vehicles capacity, 3500 liters for each vehicle
- **D_i** : demand for cities i
- **c_{ij}** : cost from city i to city j
- **k**: production capacity, 8000 liters for month
- **β** : beta $\beta = 0, \dots, 1$
- **p_w** : price per liter in each scenario
- **π_w** : probability for each scenario
- **f**: cost of the production ($\approx 30\text{€}$)
- **P_c** : price in the first stage

MATHEMATICAL FORMULATION

VARIABLES:

- Y_{ijv}
 - $Y_{ijv} = 0$ vehicle not used
 - $Y_{ijv} = 1$ vehicle used
- Q_c : quantity produced in the first stage
- Q_w : quantity produced for each scenario
- ϵ_w : ksi, variable for the CVaR
- η : eta, variable for the CvaR

OBJECTIVE FUNCTION

$\max_{Q_c, Q_w}$

$$(1 - \beta) \times \left(\left(p_c \times Q_c + \sum_{w=1}^3 \pi_w \times (p_w \times Q_w) \right) - \left(\left(Q_c + \sum_{w=1}^3 \pi_w \times Q_w \right) \times f + \sum_{i,j} c_{i,j} \right) \right) \\ - \beta \times \left(\eta - \left(\frac{1}{0.1} \right) \times \sum_{w=1}^3 \pi_w \times \varepsilon_w \right)$$

CONSTRAINTS

POSITIVITY

$$Q_c, Q_1, Q_2, Q_3 \geq 0$$

$$\varepsilon_w \geq 0$$

PRODUCTION

$$Q_c + Q_w \leq k, \forall w$$

$$Q_c + Q_w \geq \sum_i D_i, \forall w$$

CONSTRAINTS

ROUTES

$$y_{17}^{v1} = 1 \quad y_{76}^{v1} = 1 \quad y_{65}^{v1} = 1 \quad y_{54}^{v1} = 1 \quad y_{41}^{v1} = 1$$

$$y_{12}^{v2} = 1 \quad y_{211}^{v2} = 1 \quad y_{1110}^{v2} = 1 \quad y_{109}^{v2} = 1 \quad y_{98}^{v2} = 1 \quad y_{81}^{v2} = 1$$

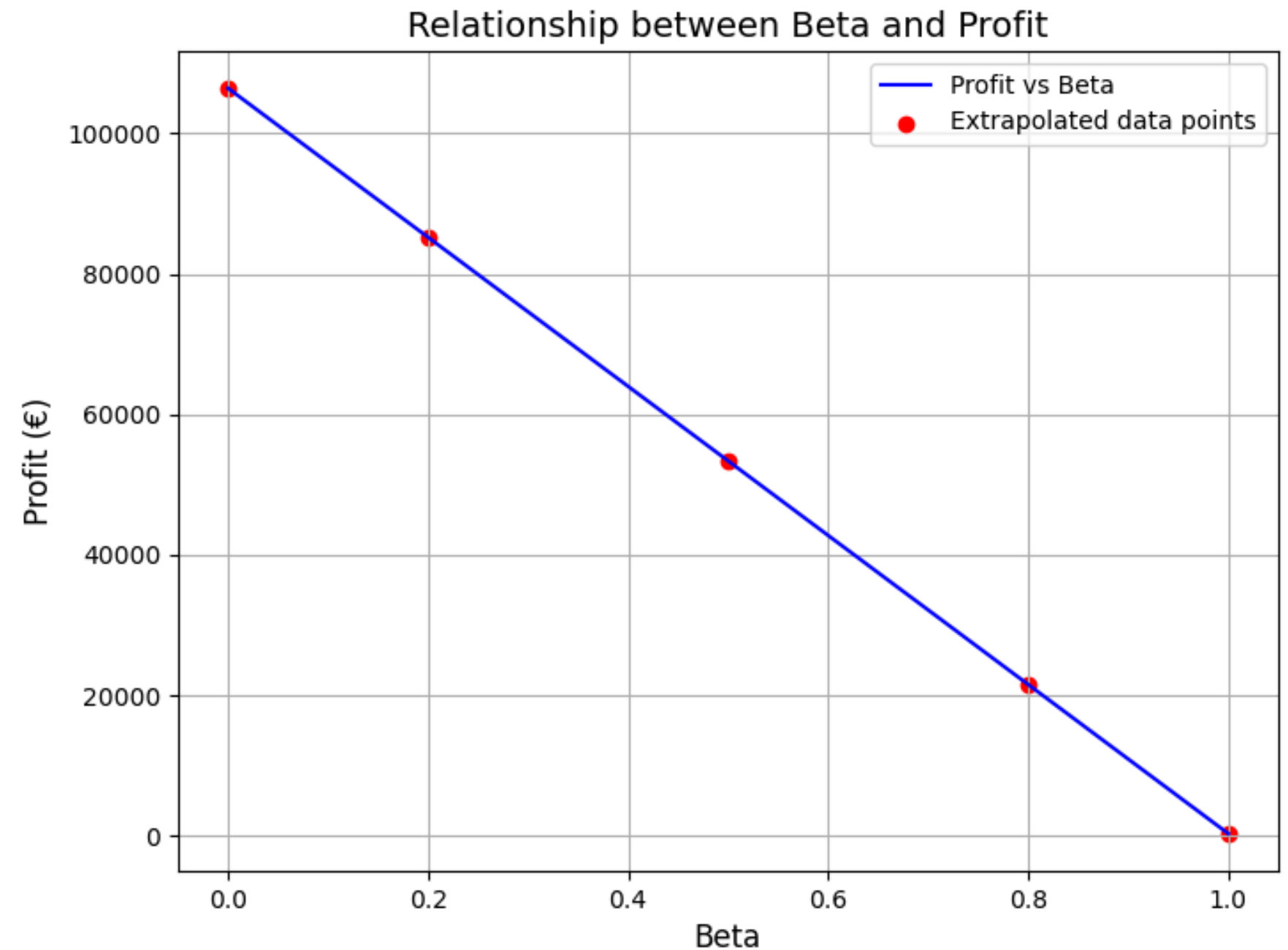
$$y_{13}^{v3} = 1 \quad y_{31}^{v3} = 1$$

CVAR

$$\eta - (Q_c + \pi_w * Q_w) * f - \sum_{i,j} c_{ij} \leq \varepsilon_w, \forall_w$$

RISK ANALYSIS - CVAR

Using CVaR for profit maximization focuses on increasing expected profit in worst-case scenarios by targeting the lower end of the profit distribution beyond a set confidence level . This helps manage risks from price changes caused by market conditions, demand shifts, or geopolitical factors ect.



RISK ANALYSIS - CVAR

$$\eta - (Q_c + \pi_w * Q_w) * f - \sum_{i,j} c_{ij} \leq \varepsilon_w, \forall_w$$

Eta represents the minimum acceptable profit level across the scenarios considered in the Conditional Value at Risk (CVaR) model.

A value of 349,000 indicates that the company aims to ensure profits remain at or above this level, even under the most unfavorable scenario.

RISK ANALYSIS - CVAR

The inverse relationship between price and CVaR suggests that as prices increase, the expected losses (or risks) in adverse scenarios become smaller. This may imply that higher prices improve the profit margin or reduce cost-related risks.

	Price	CVaR
Scenario 1	43.5	121061
Scenario 2	46.4	57161
Scenario 3	48.2	33731

	Profit
$\beta = 0.2$	85049.80
$\beta = 0.5$	53025.25
$\beta = 0.9$	10325.85

RESULTS

The decision to produce only in the scenarios and not in the first stage reflects a strategic approach to **minimize risk under uncertainty**.

Producing 7100 liters in the scenarios ensures that production is tailored to meet the demand across cities while staying within the company's production capacity (8000 liters per month).

Not producing in the first stage avoids the risk of producing at a potentially unprofitable price (e.g., 30€/l cost vs. uncertain revenue).

VARIABLE Q_first_stage.L = 0.000

VARIABLE Q.L quantity produced for each scenario

w1 7100.000, w2 7100.000, w3 7100.000

RESULTS

The model ensures a balance between maximizing profit and controlling costs. Profits vary across scenarios but always exceed the eta threshold of €349,000.

The highest profit was achieved in the scenario with the olive oil price of €48.2/l, emphasizing the importance of favorable market conditions.

Despite the capacity of 8000 liters, the optimized production is capped at 7100 liters, likely due to transportation and cost constraints.

Highlight the trade-off between demand and vehicle limitations.

CONCLUSIONS

The model demonstrates how decision theory and multi-criteria analysis can support strategic decision-making in uncertain environments.

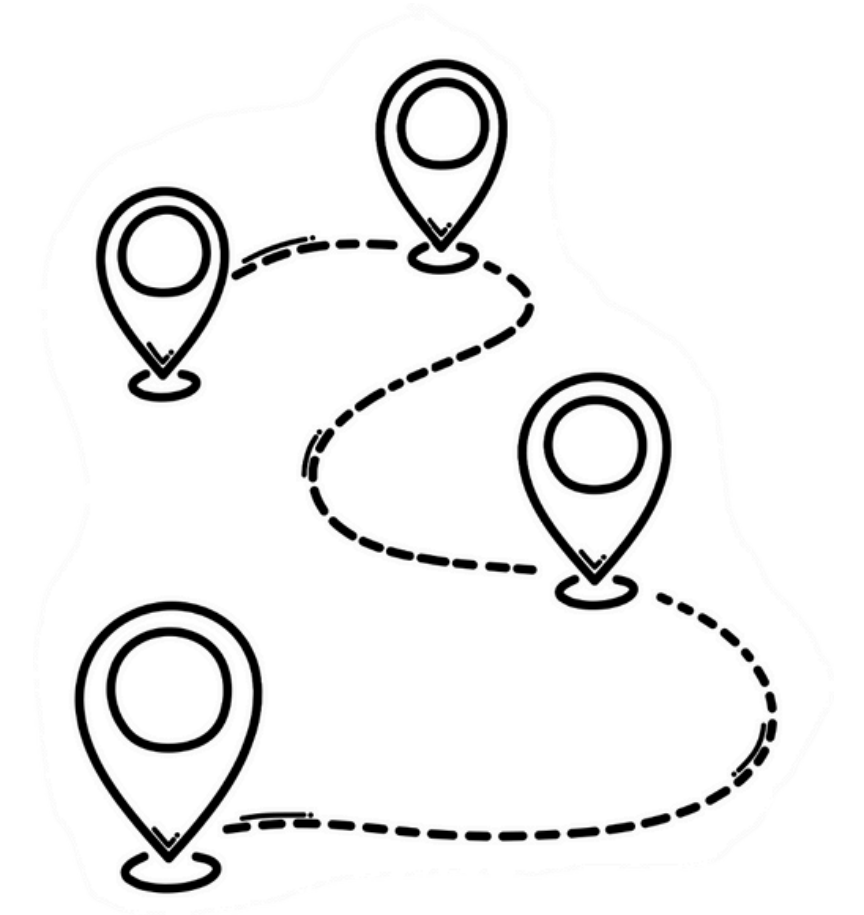
The optimal solution ensures that production meets demand while **maximizing profits under uncertain pricing conditions**.

Scenario Comparison:

1. Scenario with the highest oil price (48.2€/l) yielded the highest profit, but it carries the risk of overestimating future prices.
2. Lower price scenarios (e.g., 43.5€/l) provide insights into maintaining operational stability under adverse conditions.

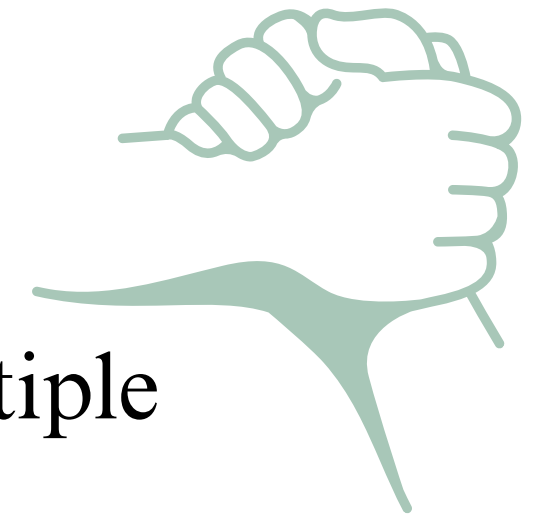
CONCLUSIONS

The **routes** are optimized to minimize the total travel distance while ensuring all demand is met. Each vehicle travels a specific subset of cities, ensuring balanced and efficient deliveries.



The model is **robust** as it effectively handles price uncertainties, manages risk through CVaR constraints, and consistently delivers cost-efficient and feasible solutions across all scenarios.

SKILLS AND WHAT WE LEARN



HARD SKILLS:

- Mathematical modeling of complex problems such as: balancing multiple constraints like cost, demand, and production capacity.
- Applying Conditional Value at Risk (CVaR) to account for uncertainty and manage risk effectively, in a maximization problem.
- Gained proficiency in using optimization software (GAMS) for solving large-scale problems efficiently.

SOFT SKILLS:

- Worked collaboratively to divide tasks, share ideas, and troubleshoot challenges during model development.
- Sharing knowledge to enhance collective understanding and project outcomes.

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